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Research Interests

Research Area: Artificial Intelligence, Embodied AI, World Models, Vision-Language-Action (VLA) Model, Multi-Agent Systems. We introduce **POInfra**, a research framework for policy optimization with generative models for robot control and multi-agent decision-making.

Education & Work Experience



National University of Singapore

Visiting PhD Student

2026–Present



Sun Yat-Sen University – Phd

Computer Science and Technology

2024–Present



Huawei Technologies Co., Ltd.

Hardware Engineer, Consumer Business Group

2023–2024



Sun Yat-Sen University – Master

Electronic Science and Technology

2020–2023



China Agricultural University – Bachelor

Electronic Information Engineering

2016–2020

Selected Publications & Preprints

MA-WAM: Multi-Agent World-Action Model for Test-Time Planning 2026

Guowei Zou, H. Wang, G. Wang, B. Zhang, Z. Chen, G. Wang, H. Wu. **Under review**

G2MAF: Test-Time Gradient Guidance for Multi-Agent Flow Policies 2026

Guowei Zou, H. Wang, G. Wang, Z. Chen, B. Zhang, G. Wang, H. Wu. **Under review**

Co- π -tree: Distilling LLM Reasoning into an Interpretable Policy Tree for Human-AI Collaboration 2026

B. Zhang, Y. Liang, **Guowei Zou**, H. Wang, H. Wu. **EMNLP 2026**

CoFlow: Coordinated Few-Step Flow for Offline Multi-Agent Decision Making 2026

Guowei Zou, H. Wang, Beiwen Zhang, Boning Zhang, H. Wu. **arXiv:2605.01457, 2026**

One Step Is Enough: Dispersive MeanFlow Policy Optimization (DMPO) 2026

Guowei Zou, H. Wang, H. Wu, Y. Qian, Y. Wang, W. Li. **ACM Multimedia (ACM MM), 2026**

PACT: Phenotype-Aware Contrastive Team Representation for Multi-Phenotype Grouped Ad Hoc Teamwork 2025

B. Zhang, Y. Liang, **Guowei Zou**, H. Wang, C. Liu, H. Wu. **arXiv:2510.25340, 2025**

DM1: MeanFlow with Dispersive Regularization for 1-Step Robotic Manipulation 2025

Guowei Zou, H. Wang, H. Wu, Y. Qian, Y. Wang, W. Li. **arXiv:2510.07865, 2025**

D²PPO: Diffusion Policy Policy Optimization with Dispersive Loss 2025

Guowei Zou, W. Li, H. Wu, Y. Qian, Y. Wang, H. Wang. **AAAI 2026**

Additional Publications

Modeling light propagation for under-display sensing in a smartphone Q. Wang, Y. Liu, Guowei Zou , Z. Liang, et al. Optics Express	2025
18-3: Mini-LED LCDs Integrated with High-capacity MIMO Visible Light Communication Z. Zhao, Y. Qiu, Guowei Zou , J. Weng, B.-R. Yang, Z. Qin. SID Symposium Digest of Technical Papers	2024
High-capacity MIMO visible light communication integrated into mini-LED LCDs Z. Zhao, Y. Qiu, Guowei Zou , Y. Liu, et al. Optics Express	2024
Enhancing the spatial resolution of light-field displays without losing angular resolution by a computational subpixel realignment W. Yang, Y. Cheng, Guowei Zou , B.-R. Yang, Z. Qin. Optics Letters	2023
Two-field sequential color liquid crystal displays with deep learning-enabled real-time driving Z. Wang, Guowei Zou , Y. Li, C. Ma, et al. Optics Letters	2023
Deep learning-based real-time driving for 3-field sequential color displays with low color breakup and high fidelity Z. Wang, Guowei Zou , Y. Shen, B.-R. Yang, Z. Qin. Optics Express	2023
65-1: Deep Learning-Enabled Image Content Adaptive Driving Algorithm for Field Sequential Color LCDs with Mini-LED Backlight Guowei Zou , Z. Wang, W. Yang, B.-R. Yang, Z. Qin. SID Symposium Digest of Technical Papers	2022
42-2: Spatial resolution-tripled integral imaging light field displays with no loss of angular resolution by recombining subpixels with zero sampling error W. Yang, Y. Cheng, Guowei Zou , B.-R. Yang, Z. Qin. SID Symposium Digest of Technical Papers	2022
Deep learning-enabled image content-adaptive field sequential color LCDs with mini-LED backlight Guowei Zou , Z. Wang, Y. Liu, J. Li, et al. Optics Express	2022

Granted Patents

Visible Light Communication Method, Device and System Using Field Sequential Color LCD. Z. Qin, Z. Zhao, Guowei Zou , Y. Qiu, B.-R. Yang. ZL202211458797.0	Granted 2026
Adaptive Refresh Rate Field Sequential Color Display. Z. Qin, Guowei Zou , Z. Wang, H. Liang, B.-R. Yang. ZL202210380314.3	Granted 2025
Deep Learning-based Control for Field Sequential Display. Z. Qin, Guowei Zou , Q. Luo, W. Yang, Z. Qiu, Z. Wu, B.-R. Yang. ZL202110793074.5	Granted 2023
High-resolution Light Field Display Method. Z. Qin, W. Yang, Y. Cheng, Guowei Zou , Y. Gong, Z. Wu, B.-R. Yang. ZL202110873615.5	Granted 2022

Internship Experience

 China Star Optoelectronics Technology Co., Ltd. (CSOT) 2022–2023 AI for image enhancement: computer-vision-based modeling and optimization for image quality improvement.	
 Huawei Technologies Co., Ltd. 2021–2023 VR/AR image enhancement: AI-based image restoration and visual quality improvement for AR-HUD systems.	
 Vivo Mobile Communication Co., Ltd. 2021–2023 Mobile vision sensing: deep learning-based sensor modeling and signal enhancement for mobile imaging.	
 Tridium Inc. (Honeywell) 2019–2020 AIoT smart building: sensor data analysis and data-driven automation for smart building systems.	
 Wlkata Robotics 2019–2020 Robotic arm control: motion planning and control algorithms for robotic manipulation.	

Honors & Awards

Distinguished Student Paper Award International Conference on Display Technology (ICDT), Hefei, China <i>Simulation methods for under-display sensing: Near-field and far-field light propagation.</i>	2024
Distinguished Student Paper Award International Conference on Display Technology (ICDT), Nanjing, China <i>Mini-LED LCDs Integrated with High-capacity MIMO Visible Light Communication.</i>	2023
Distinguished Student Paper Award International Conference on Display Technology (ICDT), Fuzhou, China <i>Simulation and analysis of edge ghosting for microcapsule e-paper based on particle dynamics and light scattering model.</i>	2022
Best Paper Award Cross-Strait Postgraduate Workshop on Display Technology (PGWS), Hong Kong, China	2021
Beijing Excellent Undergraduate Thesis Beijing Municipal Government Top 1%	2020
Excellent Undergraduate Thesis China Agricultural University Top 3%	2020

Selected Conference Presentations

The 34th ACM International Conference on Multimedia (ACM MM 2026) Rio de Janeiro, Brazil Upcoming	2026
AAAI Conference on Artificial Intelligence (AAAI-26) Singapore	2026
Society for Information Display (SID) Display Week California, USA Oral presentation	2022
International Conference on Display Technology (ICDT) Fuzhou, China Oral presentation	2022
International Display Manufacturing Conference (IDMC) Taipei, China Poster presentation	2022
Cross-Strait Postgraduate Workshop on Display Technology (PGWS) Hong Kong, China Oral presentation	2021